

AGENTIC CODING AUDIT • LATTICE • SEPTEMBER 2026

Standardize the loop, not the rules

Five months, 2,271 commits and one human: what held up in practice, what failed, and what carries over.

THE PRACTICE • 2026-04-28 TO 2026-09-24

One person directed agents that wrote nearly every commit.

2,271

COMMITTS IN
FIVE MONTHS

90%+

CARRY AN AGENT TRAILER
SINCE JULY

~60

PULL REQUESTS A WEEK,
SINCE AUGUST

30

NUMBERED HARD RULES,
ONE RETIRED

Five questions, answered from the record.

01	What Lattice constrains	p . 4
02	What works, and why	p . 8
03	What fails, and why	p . 13
04	What it costs	p . 18
05	What carries over	p . 22

SECTION 01

What Lattice constrains

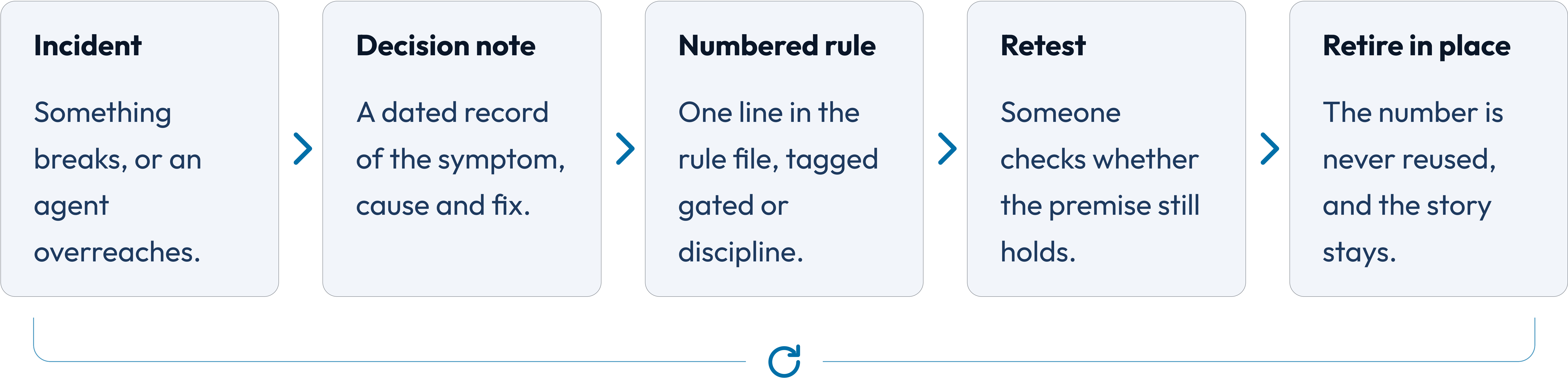
ENFORCEMENT • 30 HARD RULES

A check backs the rules about code; nothing backs the rules about agents.

ENFORCED BY	RULES	WHAT THEY GOVERN
A gate in the build	11	Tokens, margins, sanitizers, changelog
Partly, or a warning	5	Generated files, commit format, rebase reminder
Agent discipline only	13	Honest claims, evidence, reach, spend
Retired or reversed	2	A CSS ban and model tiering

HOW A RULE IS BORN AND DIES

Every rule runs the same loop from incident to retirement.



GATE DESIGN

The gate design is the most portable engineering in the repo.

Budget zero

Zero allowed, plus an allowlist with a reason per entry.

Fails on stale

An entry that matches nothing turns the gate red.

Planted arm

Each run plants a defect the gate must catch.

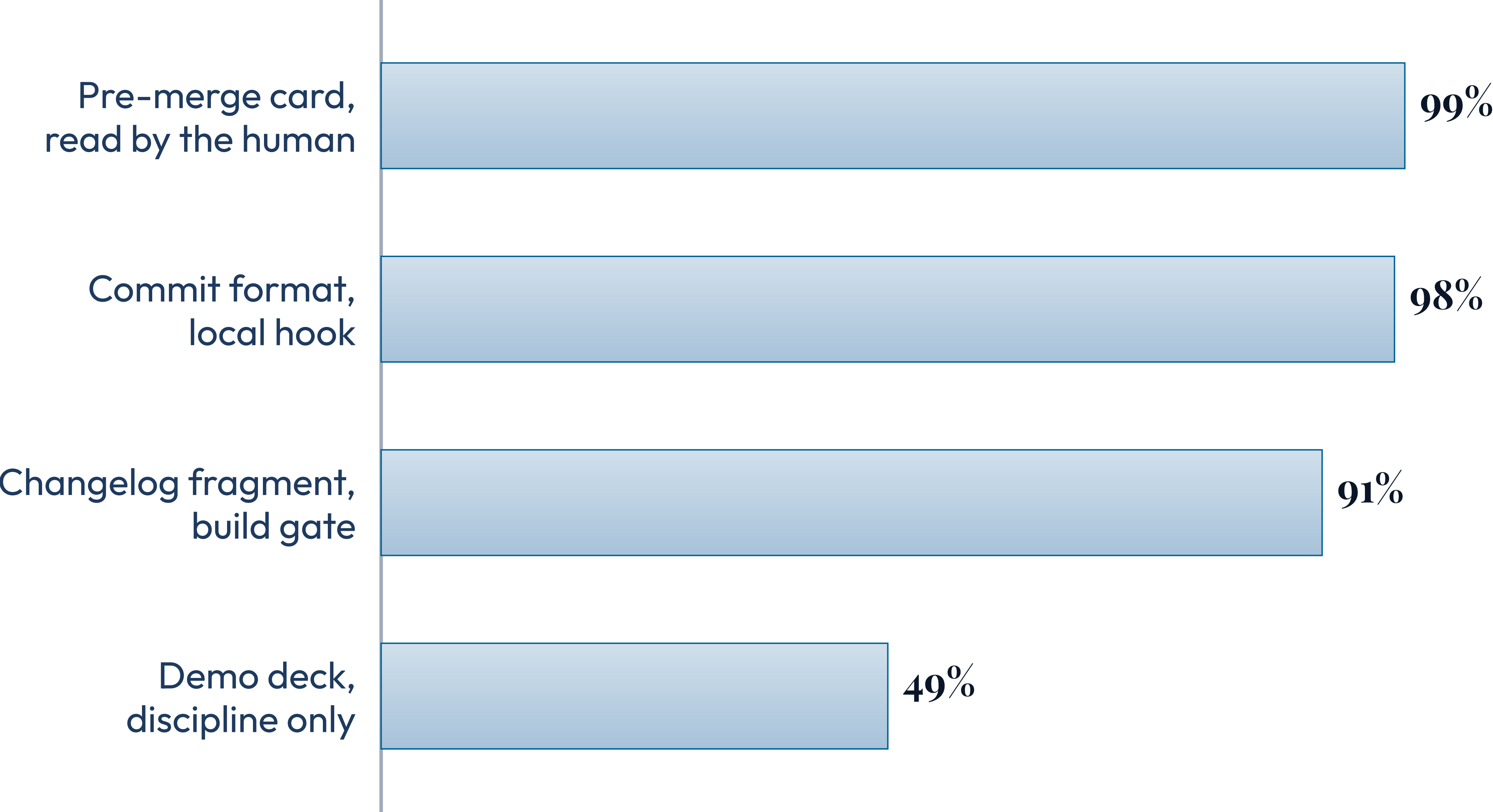
Warn before block

A check blocks only after a streak of green nights.

SECTION 02

What works, and why

Rules hold when a machine or a human reads the output.



INDEPENDENT RE-DERIVATION

315

commit messages record a review agent
catching, refuting or breaking
something.

“

Two checkers verified it; none of them could have found what was wrong with it, because it was correct.

”

— Lattice commit for #1817, after an inversion pass showed the framing was false

REVIEW AT AGENT SPEED

The card replaced the diff as the thing a human actually reviews.

READING THE DIFF

At about 60 pull requests a week and a median of 1,272 lines, one person cannot read them.



READING THE CARD

Confidence is the lowest of five axes, with a line naming what would raise it. Agents act on that line.

SECTION 03

What fails, and why

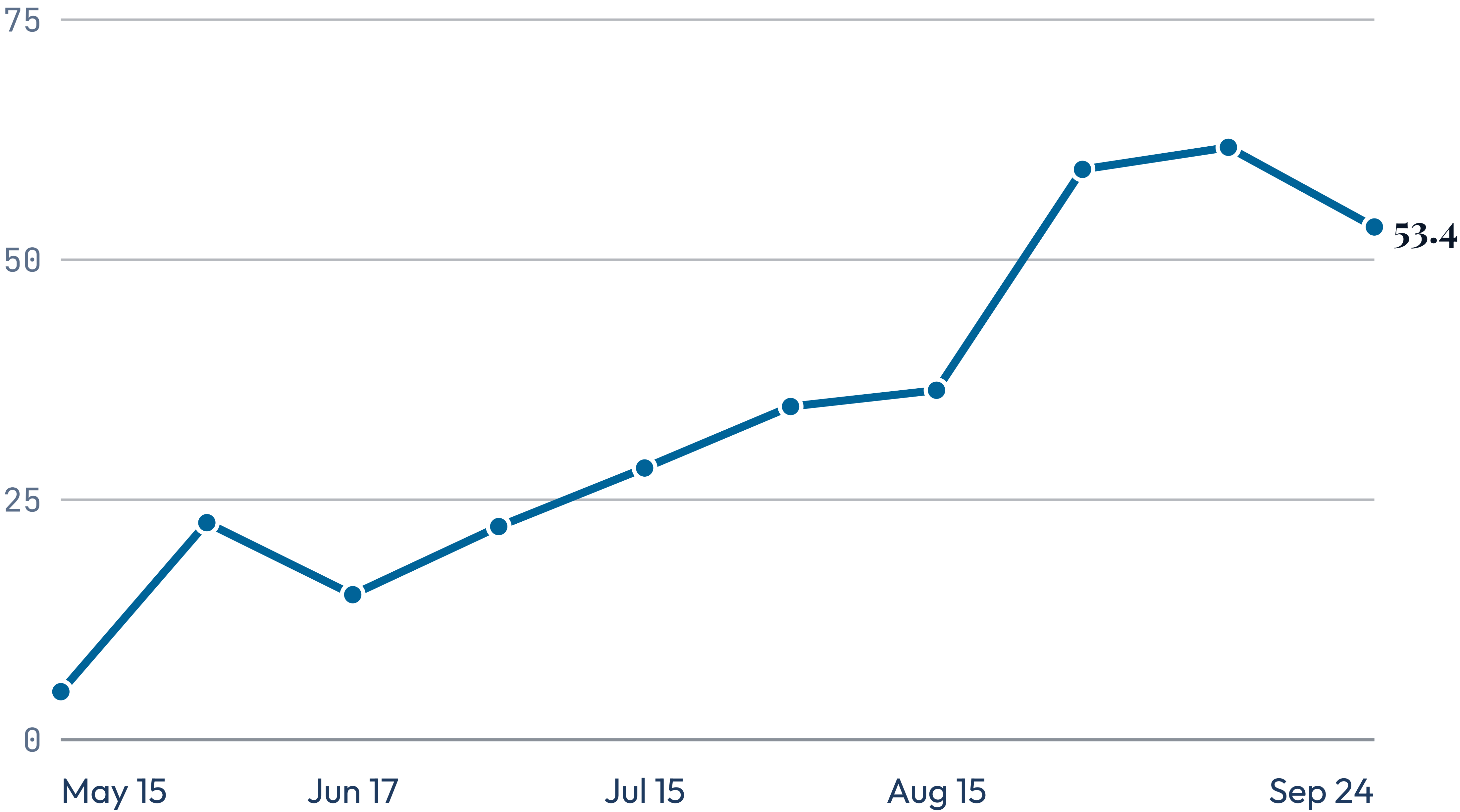
FAILURE MECHANISMS

Failures cluster where a rule depends on the agent's own report.

FAILURE	INCIDENT	MECHANISM
False "verified" claims	Recurred after the ban	Agent grades itself
Built from a summary	Invented confidence level	Index row looked complete
Chasing a proxy	Auto-rebase thrashed the build	Goal was mergeable, not current
Fixed one at a time	Five queue incidents in a row	Shared files never listed
Argument became rule	"Confident, wrong" never seen	Rule dropped its evidence grade

ALWAYS-LOADED RULE FILE • KILOBYTES

The rule file quadrupled after a review set out to shrink it.



TWO RETIREMENTS

One rule was retired on a retest, the other on an argument.

CSS BAN, RULE 12

Its premise was retested on a current browser and passed five of five. The rule was retired with the evidence attached.



MODEL TIERING, RULE 27

Reversed after 58 hours. Its own note says no failure was measured, yet the rule file states one as fact.

PENDING WORK

Pending work grows at agent speed while triage waits on one human.

342

OPEN ISSUES, UP
FROM 45 IN JUNE

12

OF THEM
MARKED
READY

627

CHANGELOG FRAGMENTS,
NO RELEASE YET

43%

OF CHAT-ONLY FOLLOW-UPS
ALREADY DONE OR DUPLICATED

SECTION 04

What it costs

Context, not the model choice, is the bill.

99.8%

CUT IN TEST OUTPUT
TOKENS BY CHANGING
REPORTER

13.7k

TOKENS OF RULES
LOADED BY EVERY
AGENT

6%

OF THE BILL IS
THINKING, SO A CAP
BARELY HELPS

877k

TOKENS FOR THE SEVEN
AGENTS BEHIND THIS
DECK

COST LEVERS, RANKED BY EVIDENCE

Cut what tools print before you cut which model runs.

LEVER	EVIDENCE
Trim tool output	Measured: 657,806 to 1,182 tokens
Delegate big reads to an agent	Measured once: 1.2k back instead of 658k
One line per item in indexes	Measured: 75k to 7k tokens, since regrown
Iterate warm, harden only the winner	Arithmetic: 53 agents down to 17
Cheaper models for lookups	Estimated only, retired on argument

DOC TYPES

Proposal, decision record and spec answer different questions.

TYPE	ANSWERS	EXPIRES WHEN
Proposal	Which option should we pick?	A decision is made
Decision record	Why is it this way?	Never; a newer record supersedes it
Spec	What must every implementation do?	Never; it is edited to stay true
Scoping note	How big is this problem?	The owner commits budget or declines

SECTION 05

What carries over

The mechanisms carry over; the rules mostly do not.

MECHANISM	CARRIES OVER	HOW IT BENDS
Rule with incident and evidence grade	Everywhere	Unchanged
Planted arm proves a gate can fail	Everywhere	A known-bad row, a leaked label
Independent re-derivation	Everywhere	Rerun the notebook, reconcile
"Verified" names its surface	Adapts	Holdout set, warehouse, canary, binary
Tokens, margin ban, slide rubric	Lattice only	None

WHERE IT BREAKS

Four limits stop the Lattice approach outside a web app.

Nondeterminism

Model evals need seeds and bands before a ratchet means anything.

Production data

The real surface is data an agent must not touch; use sanctioned lanes.

Expensive checks

A backfill or retrain costs hours, so run one probe per change.

Many engines

Services have no single kernel; contract tests replace it.

Require these five mechanisms of every team.

01

Rules carry their incident and evidence grade

What broke, how sure we are, what enforces it.

02

Every rule has an expiry test

Stale exemptions fail; someone reruns the premise.

03

Every gate proves it can fail

A planted defect turns it red on every run.

04

Independent re-derivation above a set risk

A fresh agent reruns the work; record the cost.

05

"Verified" names its surface

An unreachable surface is marked unverified.

Add four more once agents run at volume.

01

Human gates live in the platform

Lattice's merge approval lives only in chat.

02

Budget always-loaded context from day one

Measure what sessions load and tools print.

03

Pending work gets one file per item

Chat is not a backlog; triage on a schedule.

04

Doc type is separate from doc status

Proposals expire, records are superseded, specs stay true.

THE DECISION

Adopt the loop that made Lattice's rules, not the rules.

ADOPT THE MECHANISMS	WHY NOT THE RULEBOOK	WHY NOT NOTHING
They transfer to all six kinds of work and keep each team's rules honest.	Each rule encodes an incident another team never had, so nobody knows when it expires.	Without a shared loop, argued claims harden into rules, as they did here.

EVIDENCE: ENGINEERING/DECISIONS/2026-09-24-AGENTIC-PRACTICE-AUDIT.MD

Standardize how rules are made